Mohammad Khamaysi

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Professional Summary

Digital ASIC design engineer with **two tapeouts in TSMC 65 nm** and transistor-level mixed-signal design experience (flash ADC, OTA). Owned the full RTL-to-GDSII flow of two interpolation-filter chips for a QAM-64 transmitter, from the MATLAB golden model through SystemVerilog, synthesis, place and route, multi-corner sign-off, DRC/LVS and final GDS, with a bit-exact compare against the model at every stage. Strong in low-power arithmetic, timing closure and verification. Open to digital design, physical design and DSP hardware roles, and to doctoral research on energy-efficient signal-processing silicon.

Education

M.Sc. Electrical and Electronic Engineering — Tel Aviv University (2024–Present) | GPA: 93
Excellence Scholarship Recipient, 2026. Supervisor: Dr. Nimrod Ginzberg.

Thesis: Low-Power Arithmetic for Configurable Multi-Rate Digital Interpolation Filters, from MATLAB to GDSII
Designed, verified and signed off two interpolation-filter ASICs in TSMC 65 nm LP, submitted for fabrication in September 2026 through a multi-project wafer. Each 0.96 × 0.96 mm chip carries three interpolators (polyphase FIR, cubic, Hermite spline), a control unit and a cleanup filter, runs at up to 960 MHz and has seven run-time modes (three word lengths, four interpolation factors). Chip 1 loads its filter coefficients over a serial bus; chip 2 hardwires every coefficient, and its polyphase and cleanup filters are built entirely from shift-add trees with no multiplier.

- **Full ASIC flow:** Design Compiler synthesis with RTL-driven clock gating (78 gates on chip 1, 73 % of its registers; 34 on chip 2, 64 %); Innovus floorplan, power grid, placement, CCOpt clock tree and routing; Voltus power and IR drop; PrimeTime three-corner sign-off; Conformal equivalence; Calibre DRC, LVS and ERC; bond pads, dummy fill and final GDS in Virtuoso
- **Multiplier-free arithmetic:** replacing the multipliers by signed-power-of-two shift-add cut the polyphase core area by 40 % and its power by about 40 % at the same EVM (16.6 to 9.8 mW at 960 MHz, interpolation by 4); across all three cores chip 2 draws 39–60 % less core power and uses 20 % less cell area on the same die
- **Bit-exact verification:** one golden fixed-point MATLAB model compared word for word at RTL, gate, pad-ring and post-layout SDF level; 18 of 18 post-layout simulations bit-exact at 720, 900 and 960 MHz at both corners; block tests, functional coverage and VC Formal proofs
- **Timing closure at 960 MHz:** multicycle constraints for the word-rate datapaths, clock-gating checks, hold closure at the fast corner with a 50 ps guard band and three-corner sign-off; resolved setup and hold violations through constraint and RTL changes
- **Signal quality:** EVM as low as 54 m% rms on the polyphase core, the lowest of the three at every interpolation factor, from the output streams analysed in a Keysight 89600 vector signal analyser (silicon bench measurement planned); equiripple band edges chosen per interpolation factor gave 7.7 dB more image rejection at zero hardware cost
- **Algorithm study in MATLAB:** linear, Lagrange, cubic and spline interpolation compared for EVM, replica level and arithmetic cost before committing to hardware

B.Sc. Electrical and Electronic Engineering — Tel Aviv University (2020–2024) | GPA: 84
Relevant Coursework: VLSI Verification & Synthesis (100), Mixed-Signal Circuits (99), Random Signals and Noise (95), Introduction to VLSI (94), Integrated Analog IC Design (94), Digital VLSI Design (89), Signals and Systems (89), C Programming (89), Digital Electronic Circuits (86), Electronic Devices (86), Data Structures (84), Digital Logic (82)

- **Mixed-Signal Project:** 6-bit 2 GHz differential flash ADC in Virtuoso (transmission-gate sampler, comparator bank, encoder) with INL/DNL, SNDR/ENOB and power analysis
- **Analog IC Projects:** fully differential high-speed OTA with common-mode feedback, Miller compensation and noise analysis
- **VLSI Project:** 4-bit ALU with logic, timing, area and power considerations
- **Digital Circuits Project:** D-latch and flip-flop circuits optimized for timing

Teaching Experience

- Tutor – Digital Electronic Circuits, Physics II, Introduction to VLSI, Tel Aviv University
- Teaching Assistant – Energy Conversion Laboratory, Tel Aviv University
- Supervised 4 undergraduate students in their final ASIC design projects

Technical Skills

- **Languages:** SystemVerilog, Verilog, MATLAB, Python, C, Tcl, shell scripting
- **EDA Tools:** Synopsys Design Compiler, VCS, Verdi, VC Formal, PrimeTime; Cadence Innovus, Voltus, Conformal, Virtuoso (Spectre); Siemens Calibre; Keysight 89600 VSA
- **Concepts:** RTL-to-GDSII ASIC flow, fixed-point and multi-rate DSP, multiplierless (SPT/CSD) arithmetic, clock gating and low-power design, STA and timing closure, multi-corner sign-off, power and IR analysis, equivalence checking, QAM transmitter chains

Personal

- Oud player since 2018; designs and builds electronic musical instruments
- Member of the Oriental music band, Tel Aviv University; swimming

Languages: Arabic (Native), Hebrew (Fluent), English (Professional)